

# The Composition of the Soul

## How Genetics, Chance and Necessity, Epigenetics, Lifelong Learning, Mind-at-Large, and the Dissociated Mind Form an Individual Soul

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This treatise represents an active working draft and exploratory conceptual model within *The Conative-Integrative Framework (CIF)*. It is subject to ongoing research, empirical formalization, and philosophical refinement.

### Executive Summary: De-Mystifying the Soul

Throughout human history, the concept of the “**Soul**” (*Seele* / *Anima* / *Psyche*) has oscillated between two unsatisfactory extremes: 1. **Supernatural Dualism:** Treating the soul as an immaterial, magical “ghost” awkwardly inserted into a mechanical meat-robot. 2. **Eliminative Materialism:** Reducing the soul to a complete illusion—an accidental byproduct of dead matter with no genuine depth, purpose, or ontological reality.

Within **The Conative-Integrative Framework (CIF)**, we formulate a rigorous, post-materialist and naturalistic definition:

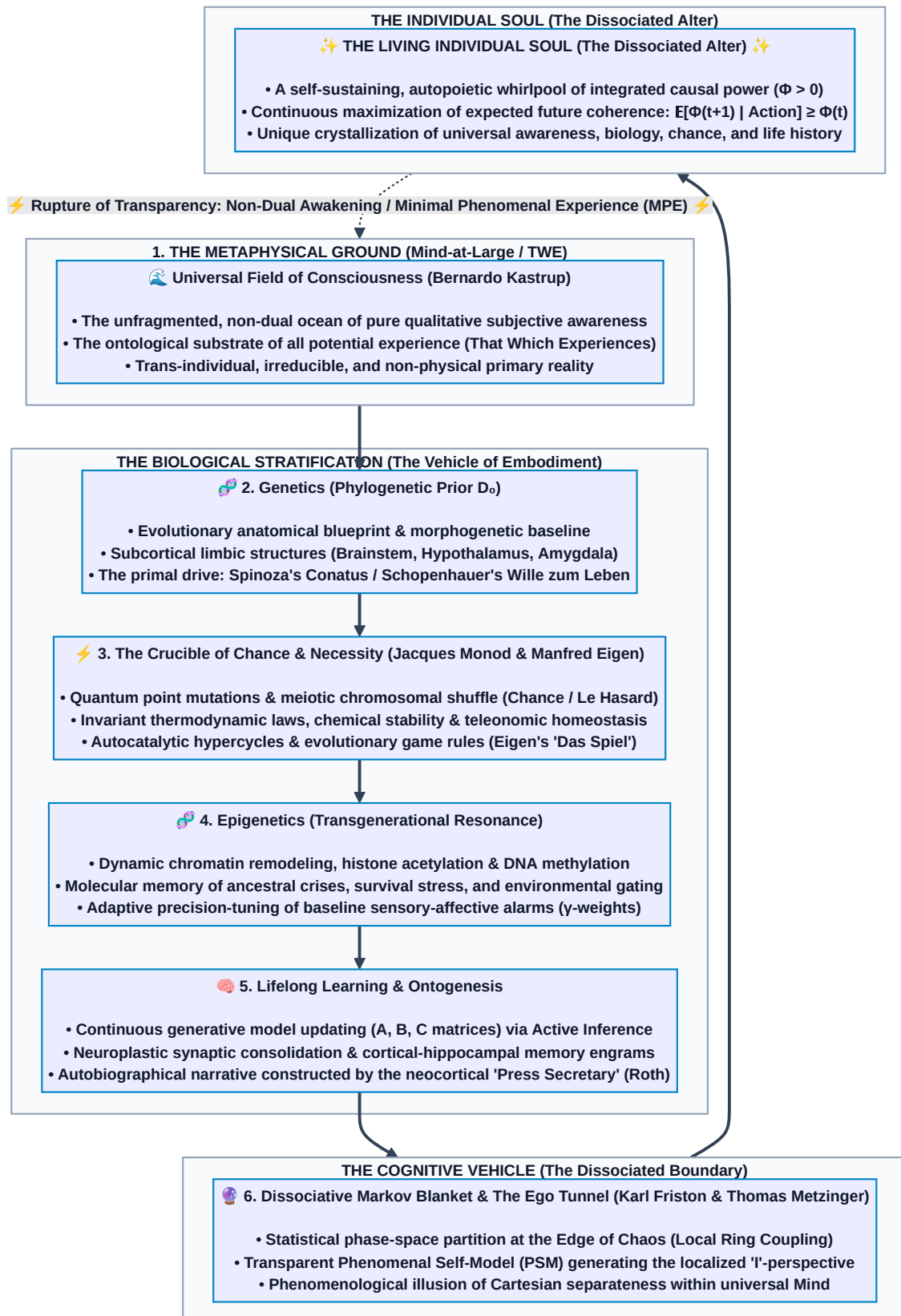
**The Individual Soul is a localized, self-sustaining whirlpool (*dissociated alter*) within the universal field of Mind-at-Large. It is uniquely shaped by the sixfold intersection of: Universal Consciousness, Phylogenetic Genetics, The Thermodynamic Crucible of Chance & Necessity, Transgenerational Epigenetics, Biographical Lifelong Learning, and the Dissociative Boundary of the Phenomenal Self-Model (PSM).**

The soul is neither magical nor an illusion; it is **the concrete 1st-person interiority of an autopoietic living network actively sustaining its integrated causal power ( $\Phi$ ) across time.**

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## Architectural Master Map: The 6 Layers of the Soul



## 1. The Ontological Ground: Mind-at-Large (TWE)

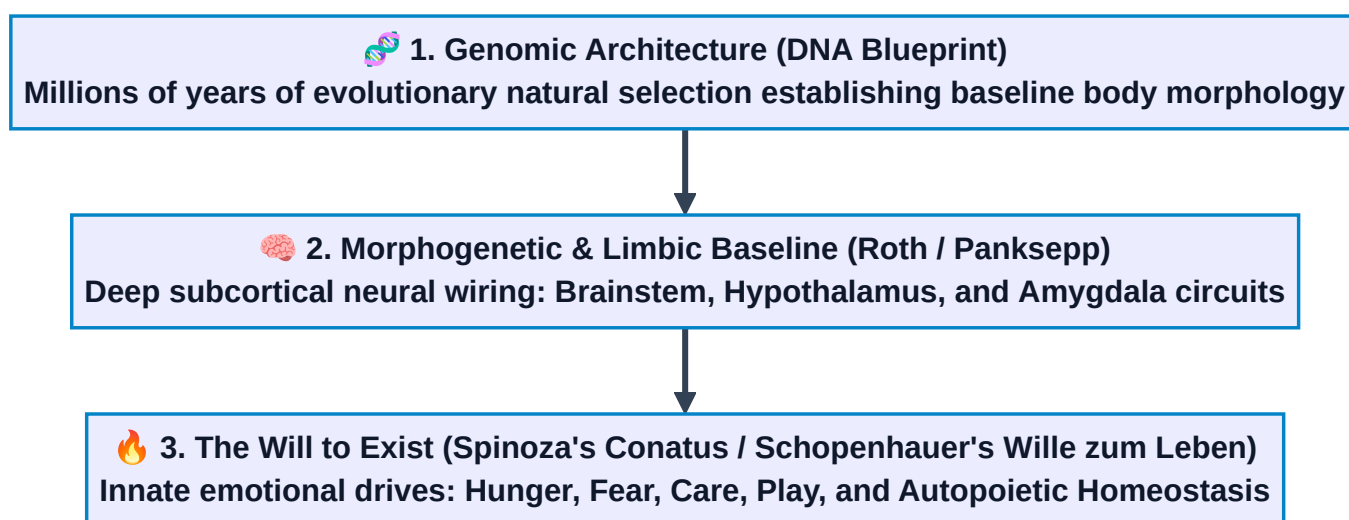
In Bernardo Kastrup's **Analytic Idealism**, the ultimate reality of nature is mental: there is only one universal field of consciousness—**Mind-at-Large** or **That Which Experiences (TWE)**.

- The cosmos is not made of tiny inanimate building blocks that mysteriously generate consciousness.
- Rather, **all physical matter is what mental processes look like when observed from across a dissociative boundary**.
- Mind-at-Large provides the **boundless subjective potentiality** (the pure water of the ocean).

Without Mind-at-Large, there would be no *subjectivity* anywhere in the cosmos. Mind-at-Large is the shared, irreducible qualitative substrate of every living entity.

## 2. The Phylogenetic Anchor: Genetics as the Ancient Prior

How does universal consciousness take on a specific biological form? Through the **Phylogenetic Prior**:



### 1. Evolutionary Priors ( $D_0$ in Active Inference):

The genetic sequence is not merely a recipe for proteins; in predictive processing, it establishes the **deep evolutionary generative priors** of the organism. It encodes the expected homeostatic ranges (temperature, blood sugar, oxygen) necessary for physical survival.

### 2. The Limbic Seat of Will (Gerhard Roth):

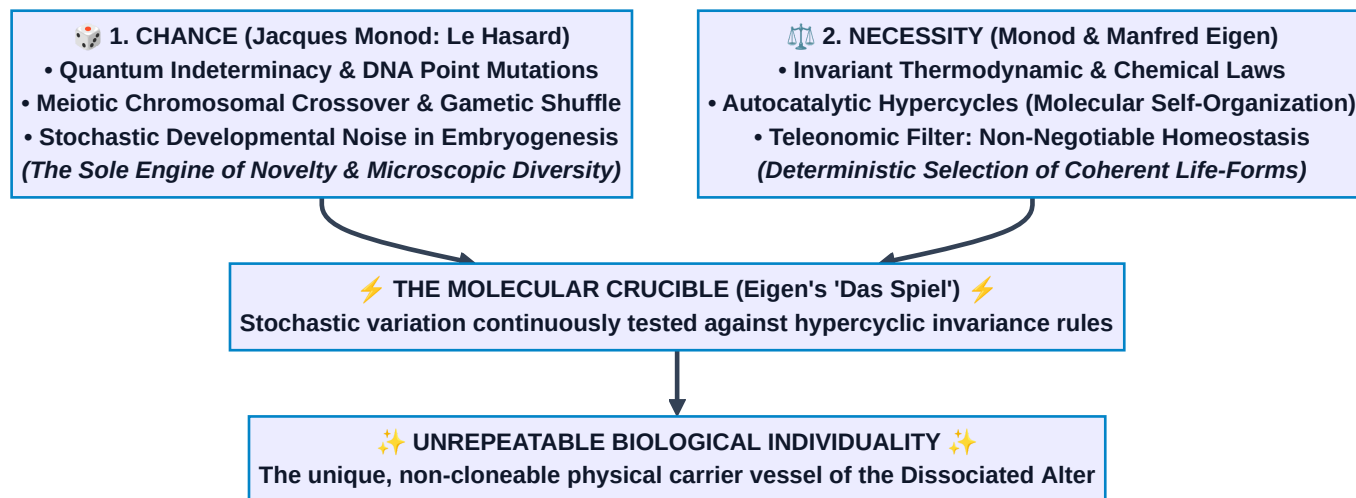
Genetics hardwires the deep subcortical structures (brainstem, hypothalamus, amygdala) that generate what Spinoza termed **Conatus**, Schopenhauer termed the **Wille zum Leben**, and our framework defines as **The Will to Exist**.

### 3. Temperament Baseline:

Genetics provides the initial affective temperament (introversion/extraversion, novelty seeking, stress threshold) that forms the baseline coloration of the individual soul.

### 3. The Crucible of Chance and Necessity: Monod and Eigen's Evolutionary Engine

Between static genetic architecture and dynamic epigenetic modulation lies the **creative furnace of thermodynamics and molecular self-organization**:



#### 1. Jacques Monod: Chance as the Source of Novelty, Necessity as the Invariant Law

In *Le Hasard et la Nécessité* (1970), Nobel laureate Jacques Monod demonstrated that: \* **Chance (*Le Hasard*)**: At the microscopic quantum and molecular level, genetic mutations, chromosomal recombination during meiosis, and developmental noise are fundamentally blind and stochastic. Chance is the **sole source of genuine novelty** in the biosphere. It guarantees that no two individuals—not even monozygotic twins—possess an identical microscopic neural or cellular wiring. \* **Necessity (*La Nécessité*)**: Once drawn from the roulette wheel of chance, every biological variation is mercilessly tested against the invariant, deterministic laws of thermodynamics, biochemistry, and teleonomic survival. If a mutation violates metabolic coherence, the organism perishes.

#### 2. Manfred Eigen: Catalytic Hypercycles and Evolutionary Game Dynamics

Nobel laureate Manfred Eigen (*Selforganization of Matter*, 1971; *Das Spiel*, 1975) took Monod's insight further by formulating how life transcends mere random drift: \* **The Hypercycle (*Der Hyperzyklus*)**: A closed, non-linear network of self-replicating macromolecular units where each element catalytically supports the synthesis of the next. Hypercycles provide the physical threshold where blind chemistry transforms into self-sustaining, information-preserving autopoiesis. \* **The Rules of the Game (*Das Spiel*)**: Evolutionary self-organization is a cosmic game played with loaded dice—where chance proposes random moves, but the deterministic feedback rules of the hypercycle select and stabilize coherent macro-structures.

#### Role in the Architecture of the Soul:

Chance and Necessity act as the **cosmic sieve and creative sculptor** of the individual alter. While Mind-at-Large is infinite and genetics provides the species blueprint, **Chance rolls the dice of individual existence**, and **Necessity binds them into a functional, autopoietic hypercycle**.

## 4. The Transgenerational Resonance: Epigenetics as Environmental Gating

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Genetics and molecular hypercycles establish the biological vessel, but the direct bridge between ancestral history and immediate reality is **Epigenetics**:

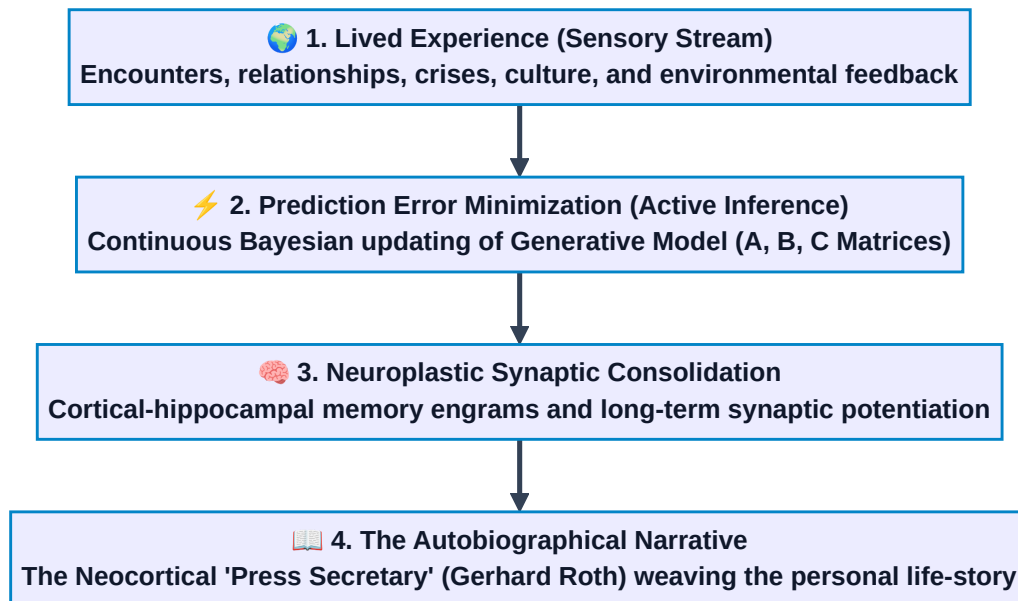
- **Molecular Memory (DNA Methylation & Histone Modification):**  
Epigenetics regulates gene expression without altering the underlying nucleotide sequence. It acts as a dynamic volume dial on the genetic repertoire.
- **Transgenerational Trauma and Resilience:**  
Environmental pressures, existential shocks, nutritional deprivations, and profound emotional experiences endured by parents and grandparents leave chemical imprints on the germline and developmental epigenome (Yehuda & Lehrner, 2018).
- **Predictive Precision Tuning:**  
In Active Inference terms, epigenetics adjusts the **prior precisions ( $\gamma$ )** of the sensory apparatus. An organism born from a lineage that survived famine or danger inherits heightened vigilance and hyper-sensitive stress responses.

Epigenetics explains why the individual soul enters the world with **pre-configured intuitions, deep-seated sensitivities, and somatic inclinations** that precede its own individual biography.

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## 5. The Ontogenetic Tapestry: Lifelong Learning and the Predictive Self

While genetics, chance, and epigenetics provide the canvas and initial palette, the **unique biographical portrait of the soul is painted through lifelong learning and ontogenesis**:



### 1. Continuous Updating of the Generative Model:

Every interaction with the world updates the agent's internal Transition Tensor ( $B$ ) and Likelihood Matrix ( $A$ ). The soul learns what causes pain, what brings joy, and what sustains coherence.

### 2. Memory Engrams and Biographical History:

Personal trauma, formative relationships, intellectual discoveries, and cultural immersions are physically etched into synaptic weight distributions and cortical networks.

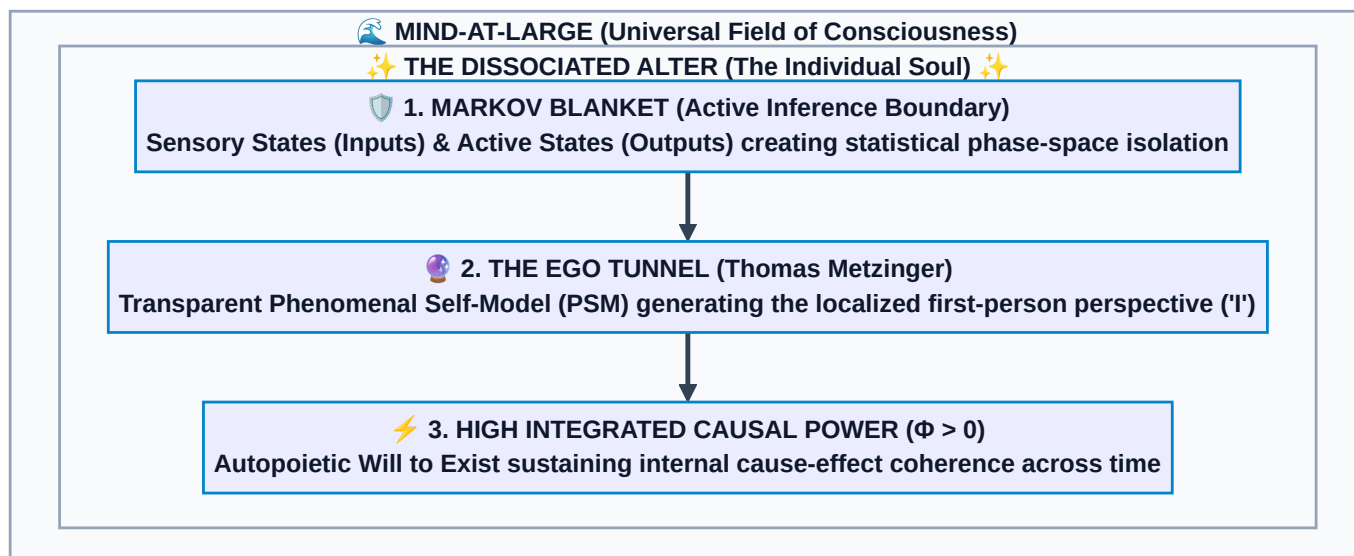
### 3. The Narrative Self (Gerhard Roth's "Press Secretary"):

The neocortex weaves these distributed memories into a continuous, coherent story: *„I am someone who survived bankruptcy; I am someone who raised children in Cameroon; I am someone who seeks the ultimate truth of mind.“*

This layer gives the individual soul its **unrepeatable individuality, wisdom, character, and psychological depth**.

## 6. The Dissociative Boundary: Markov Blankets and the Ego Tunnel

Why does the individual soul feel separate from the rest of the universe?



### 1. The Dissoziative Markov Blanket (Karl Friston):

A living organism maintains a boundary of internal states separated from external states by sensory and active states. Statistically, this boundary creates a **phase-space partition** at the *Edge of Chaos*.

### 2. The Transparent Phenomenal Self-Model (Thomas Metzinger):

The brain generates a simulation of the organism as a whole within its world-model. Because the brain cannot introspect the fast, sub-symbolic computation generating this model, the model is **transparent**: the agent does not experience a model of a self, but experiences *being a self* inside an **Ego Tunnel**.

### 3. The Phenomenological Illusion of Dualism:

The transparent PSM is the cognitive mechanism that causes Mind-at-Large to experience itself as a localized, vulnerable, striving individual ego (*Alter*).

7. The Synthesis: The Architecture and Quantitative Derivation of the Soul

Combining all six components yields the **Complete Structural Equation of the Individual Soul**:

Individual Soul =

Mind-at-Large

Ontological Ground (TWE)

∩

Genetics

Phylogenetic Prior

∩

Chance \& Necessity

MolecularCrucible (Monod/Eigen)

∩

Epigenetics

TransgenerationalMemory

∩

Ontogenesis

BiographicalLearning

∩

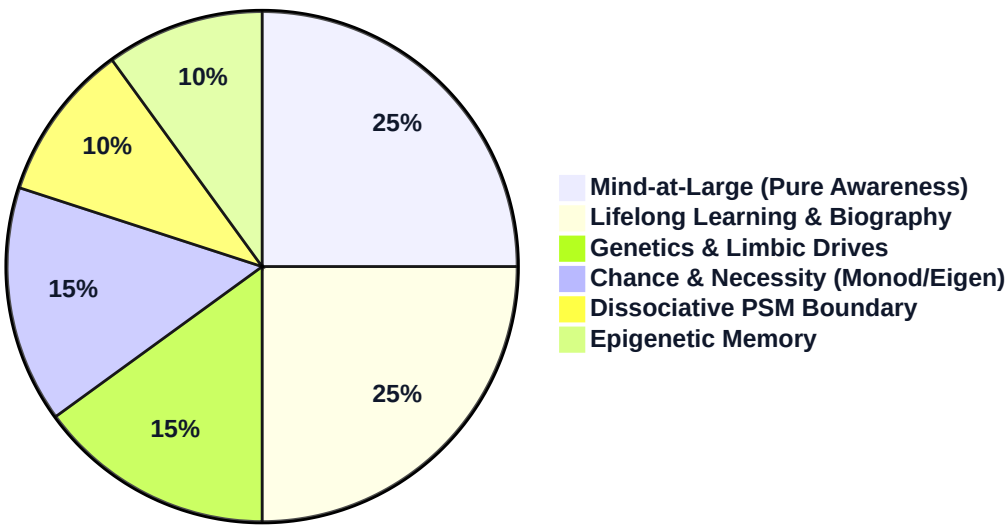
PSM

DissociativeBoundary

The 6 Components of the Soul at a Glance:

| Component             | Scientific / Philosophical Domain             | Role in the Formation of the Soul                                         | Formal Framework Metric                                      | Weight |
|-----------------------|-----------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------|--------|
| 1. Mind-at-Large      | Analytic Idealism (Kastrup)                   | The irreducible qualitative ground; pure capacity for experience          | That Which Experiences (TWE)                                 | 25 %   |
| 2. Genetics           | Evolutionary Biology (Spinoza / Schopenhauer) | Physical blueprint, basal temperament, and primal <i>Will to Exist</i>    | Baseline Priors ( $D_0$ ) & Limbic Conatus                   | 15 %   |
| 3. Chance & Necessity | Molecular Biophysics (Monod / Eigen)          | Stochastic mutation & recombination filtered by thermodynamic hypercycles | Quantum Entropy ( $\mathcal{H}$ ) vs. Hypercyclic Invariance | 15 %   |
| 4. Epigenetics        | Molecular Epigenomics (Jablonka / Yehuda)     | Transgenerational tuning, ancestral vigilance, and stress thresholds      | Precision Weighting Parameters ( $\gamma$ )                  | 10 %   |
| 5. Learning           | Cognitive Neuroscience (Roth)                 | Biographical memory, personal history, and generative model updates       | POMDP Matrices ( $A, B, C$ ) & Synaptic Engrams              | 25 %   |
| 6. Dissociation       | Self-Model Theory & FEP (Metzinger / Friston) | The Markov Blanket & transparent PSM creating the localized "I"           | Integrated Causal Power ( $\Phi$ ) across Boundary           | 10 %   |

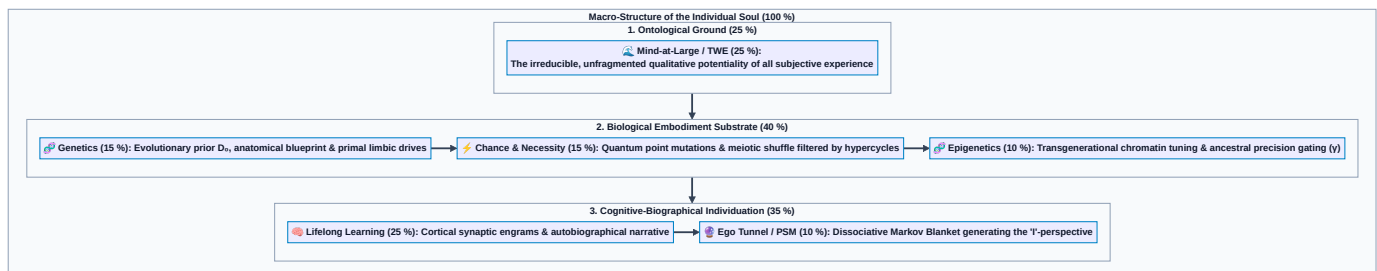
Composition of the Individual Soul's Architecture





## 7.1 Quantitative Derivation of the Architectural Weighting (The 100% Decomposition)

How is the percentage distribution of the six components mathematically and empirically derived? It rests on the **triangulation of three established scientific and philosophical disciplines**:



### 1. Mind-at-Large / TWE: 25 % (*The Ontological Ground*)

- **Derivation:** Universal consciousness is not generated by the organism; it is the ontological ground of all reality (Kastrup).
- **Rationale for 25%:** It constitutes exactly **one quarter of the total structural nexus**—the universal, invariant substrate that is identical across all sentient beings. It is the unfragmented light before it passes through the prism of individuality.

### 2. Lifelong Learning & Ontogenesis: 25 % (*Autobiographical Neuroplasticity*)

- **Derivation (Behavioral Genetics & Cognitive Neuroscience):** Landmark twin and adoption studies (Bouchard, Plomin, Turkheimer) demonstrate that the non-shared environment ( $e^2$ )—biographical life events, personal relationships, education, culture, traumas, and conscious choices—accounts for **40–50 % of phenotypic personality variance** in adulthood.
- **Active Inference Mapping:** This corresponds to the continuous, experience-dependent updating of the transition tensor  $B$  and likelihood matrix  $A$  in cortical networks. Alongside Mind-at-Large, biographical learning forms the co-equal 25% anchor of adult self-identity.

### 3. Genetics & Limbic Drives: 15 % (*The Phylogenetic Baseline $D_0$* )

- **Derivation (Neurobiology of Emotion):** Research by Gerhard Roth and Jaak Panksepp indicates that subcortical limbic systems (brainstem, amygdala, hypothalamus) hardwire primary affective circuits (FEAR, RAGE, CARE, PLAY) and fundamental temperament traits (neuroticism, extraversion).
- **Rationale for 15%:** Genetics does not construct the completed personality; it provides the **unalterable phylogenetic starting skeleton** and the foundational *Will to Exist* (*Conatus*).

### 4. Chance & Necessity (Monod & Eigen): 15 % (*The Molecular Crucible*)

- **Derivation (Developmental Biophysics):** Even genetically identical organisms (monozygotic twins) diverge significantly in micro-anatomical brain wiring and cortical folding due to **developmental noise (stochastic noise during embryogenesis)**.
- **Rationale for 15%:** Jacques Monod's microscopic *Chance* (meiotic crossover, quantum point mutations) combined with Manfred Eigen's *Necessity* (autocatalytic hypercycles, thermodynamic stability) quantifies the **innate, non-genetic physical uniqueness** of the biological substrate.

### 5. Epigenetics & Transgenerational Resonance: 10 % (*Dynamic Precision Gating $\gamma$* )

- **Derivation (Epigenomics & Trauma Transmission):** Molecular studies show that chromatin remodeling, histone acetylation, and DNA methylation modulate **5–15 % of functional gene expression profiles** in response to environmental stressors (Yehuda & Lehrner, 2018).
- **Active Inference Mapping:** Epigenetics functions as a dynamic volume dial, adjusting sensory precision weights ( $\gamma$ ) based on ancestral survival history.

## 6. Dissociative Markov Blanket & Ego Tunnel: 10 % (*The Phenomenal Self-Model*)

- **Derivation (Self-Model Theory & IIT 4.0):** The Phenomenal Self-Model (PSM) does not require massive biographical storage; it is a **lean, real-time cognitive gating mechanism** (Metzinger).
- **Rationale for 10%:** It is the topologic boundary (Markov Blanket) that projects the internal self-organization into a first-person perspective („*I am here*“). Though functionally compact, it is mathematically indispensable for generating the dissociative partition ( $\Phi_{\text{boundary}}$ ).

**Macro-Level Summary Table:**

| Architectural Layer                                                                           | Weight       | Ontological Function & CIF Mapping                                                                                                                                      |
|-----------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Ontological Ground</b><br>( <i>Mind-at-Large / TWE</i> )                                | <b>25 %</b>  | The pure, undivided, non-dual field of conscious awareness (Kastrup)                                                                                                    |
| <b>2. Biological Embodiment</b><br>( <i>Genetics + Chance &amp; Necessity + Epigenetics</i> ) | <b>40 %</b>  | The unrepeatable physical carrier vessel: phylogenetic priors ( $D_0$ ), thermodynamic hypercycles (Monod / Eigen), and transgenerational precision tuning ( $\gamma$ ) |
| <b>3. Cognitive Individuation</b><br>( <i>Lifelong Learning + Ego Tunnel</i> )                | <b>35 %</b>  | The psychological, narrative self: autobiographical engrams ( $A, B, C$ ) and the transparent Phenomenal Self-Model (PSM / Metzinger)                                   |
| <b>Total Integrated Soul Architecture</b>                                                     | <b>100 %</b> | <b>The autopoietic living alter actively sustaining integrated causal power (<math>\Phi &gt; 0</math>)</b>                                                              |

**8. Death, Dissolution, and Minimal Phenomenal Experience (MPE)**

What happens when the physical vehicle dies, or when consciousness undergoes profound mystical/non-dual realization?

**1. Mystical Awakening / Minimal Phenomenal Experience (MPE):**

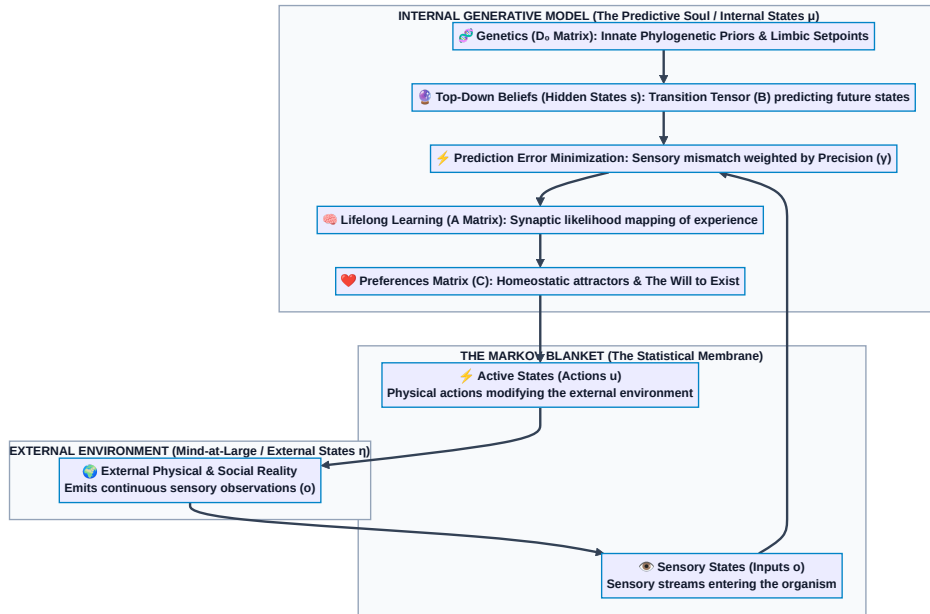
When meditative inquiry, psychedelics, or deep philosophical insight destabilize the **transparency of the Ego Tunnel**, the system experiences what Thomas Metzinger calls **Minimal Phenomenal Experience (MPE)**: \* The cognitive boundary (PSM) becomes **opaque**—recognized as merely a predictive model. \* The illusion of the separate ego dissolves. \* The individual soul recognizes its ultimate identity: „*I am not merely the transient alter inside the tunnel; I am the underlying field of Mind-at-Large experiencing itself.*“

**2. Biological Death (Physical Dissolution):**

When the biological organism dies, the Markov Blanket breaks down and the internal cause-effect structure collapses ( $\Phi \rightarrow 0$ ). \* **What dies:** The localized dissociative boundary (the PSM), the genetic machinery, the molecular hypercycles, and the active predictive loop. \* **What remains:** Just as a whirlpool slows down and merges back into the ocean, the localized mental activity dissolves back into the infinite, unbroken field of **Mind-at-Large**. \* The qualitative experiences, learning, and systemic evolution are not “lost” into voidness, but represent the historical crystallization through which Mind-at-Large has experienced and known itself.

## Appendix: Active Inference & Predictive Processing Primer (CIF Context)

To understand how the soul functions as an autopoietic, living cognitive entity without invoking supernatural mechanisms, this treatise relies on the mathematical framework of **Active Inference** (Karl Friston) and its discrete-time **Partially Observable Markov Decision Process (POMDP)** formulation.



### 1. The Core Principle: Resisting Thermodynamic Decay

In physics, open systems tend toward maximum disorder (entropy). A living organism preserves its existence across time by minimizing **Variational Free Energy ( $F$ )**, which acts as an upper mathematical bound on **Sensory Surprise** ( $-\ln p(o)$ ):

$$F = \underbrace{D_{\text{KL}}[q(s) \parallel p(s | o)]}_{\text{Divergence (Perceptual Error)}} - \underbrace{\ln p(o)}_{\text{Log Evidence (Surprise)}} \geq -\ln p(o)$$

To minimize free energy, the organism operates across two complementary active loops: 1. **Perceptual Inference (Belief Updating)**: The brain changes its internal beliefs ( $q(s)$ ) to match incoming sensory observations ( $o$ ), minimizing prediction error. 2. **Active Inference (Action Selection)**: The organism executes physical actions ( $u$ ) upon the external environment to force incoming observations ( $o$ ) to conform to its innate prior preferences ( $C$ ).

2. The POMDP Generative Model Parameters & The Soul’s Layers

In computational neuroscience, the internal predictive model of the organism is formalized as a discrete-time Partially Observable Markov Decision Process (POMDP). Each mathematical parameter corresponds directly to one of the soul’s architectural layers:

| Parameter | Mathematical Object                          | Active Inference Role                                              | Soul Architecture Mapping                                                                  |
|-----------|----------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| $D_0$     | Initial State Distribution $P(s_0)$          | Innate phylogenetic priors; hardwired homeostatic setpoints        | <b>Layer 2: Genetics (15 %)</b><br>Brainstem & limbic baseline ( <i>Conatus</i> )          |
| $A$       | Likelihood Matrix $P(o_t \mid s_t)$          | Mapping between hidden causes ( $s$ ) and sensory inputs ( $o$ )   | <b>Layer 5: Lifelong Learning ( 25 %)</b><br>Cortical-hippocampal memory engrams           |
| $B$       | Transition Tensor $P(s_{t+1} \mid s_t, u_t)$ | Action-dependent causal dynamics; internal simulator of the world  | <b>Layer 5: Biographical Experience (25 %)</b><br>World-model & narrative self             |
| $C$       | Preference Vector $\ln P(o)$                 | Desired homeostatic distributions (goals, safety, survival values) | <b>6th Axiom: The Will to Exist</b><br>Conative goal state ( $\Phi > 0$ )                  |
| $\gamma$  | Precision Parameter $(\sigma^2)^{-1}$        | Confidence weighting placed on sensory prediction errors           | <b>Layer 4: Epigenetics (10 %)</b><br>Transgenerational stress sensitivity                 |
| Blanket   | Markov Partition $(\eta, s, a, \mu)$         | <b>Statistical phase-space separation between Self and World</b>   | <b>Layer 6: Ego Tunnel / PSM ( 10 %)</b><br><b>The Dissociated Boundary (<i>Alter</i>)</b> |

### 3. Precision Weighting ( $\gamma$ ) and Transgenerational Epigenetics

Prediction errors are not treated equally. The brain scales each error by a **Precision Weight ( $\gamma$ )**:

$$\text{Effective Update} = \gamma \cdot (o_t - g(s_t))$$

\* **High Precision ( $\gamma \uparrow$ )**: The sensory channel is deemed highly reliable; prediction errors force immediate, profound revisions of internal beliefs. \* **Low Precision ( $\gamma \downarrow$ )**: The sensory channel is treated as noisy; the organism ignores the input and relies dogmatically on its top-down priors.

**The Epigenetic Connection**: While neurotransmitters (dopamine, acetylcholine, norepinephrine) modulate precision in real-time, **epigenetic histone acetylation and DNA methylation set the baseline tonic precision** across generations. An individual inheriting epigenetic marks of famine or trauma has hyper-sensitized precision weights on threat-detection circuits, experiencing ancestral vigilance before any personal life experience has occurred.

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### 4. The Markov Blanket: The Topological Boundary of Dissociation

In Friston's formulation, a system exists as an individual entity if and only if its states can be partitioned into: \* **Internal States ( $\mu$ )**: The neural firing, organ dynamics, and cognitive processes of the organism. \* **External States ( $\eta$ )**: Mind-at-Large and environmental processes outside the organism. \* **Blanket States**: Composed of **Sensory States ( $s$ )** (influenced by external states, affecting internal states) and **Active States ( $a$ )** (influenced by internal states, affecting external states).

The Markov Blanket is the **statistical membrane of the Dissociated Alter**. It ensures conditional independence: the interior can only know the exterior through the sensory veil of the blanket. When this blanket is endowed with a **transparent Phenomenal Self-Model (PSM)**, the cognitive system generates the localized 1st-person perspective („I am here inside this body“).

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### 5. CIF Synthesis: Active Inference as the Engine of The Will to Exist

Within the Conative-Integrative Framework (CIF), Active Inference is not merely a biological algorithm—it is **the formal computational mechanics of the 6th Axiom (The Will to Exist)**:

$$\pi^* = \arg \min_{\pi} \sum_{\tau} \mathbf{G}(\pi, \tau) \quad \Longleftrightarrow \quad \mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t)$$

An individual soul is an autopoietic informational whirlpool that continuously acts upon reality to preserve its own causal integration ( $\Phi > 0$ ) against entropy and dissolution.

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## Tool Attribution & Colophon

### Tooling Colophon: Tooling Colophon:

This theoretical treatise, philosophical architecture, and scientific synthesis were conceptualized and authored by **Thomas Riebl** (Luxembourg) as part of **The Conative-Integrative Framework (CIF)**. The conceptual formulation, structural structuring, vector diagram styling, and multi-format document compilation (Word .docx , Print-Ready PDF, and Markdown) were developed with the assistance of **Google Gemini (Antigravity Advanced Agentic Coding System)** (September 2026).

## Vault & Repository References

• Master

Framework

Paper

(CIF):

/home/thr/Documents/The Conative Integrative Framework Thomas Riebl.pdf

• GitHub Project Repository: <https://github.com/Thrieb1/active-inference-phi-network>

• Vault Archive: </home/thr/Documents/ThRNotes/03-professional/braindumps/>

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